

Linear Algebra – MAT 2610

Section 6.2 (Orthogonal Sets)

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Definition

A set of vectors $\{u_1, \dots, u_p\}$ in R^n is an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, or whenever $i \neq j$

$$u_i \circ u_j = 0$$

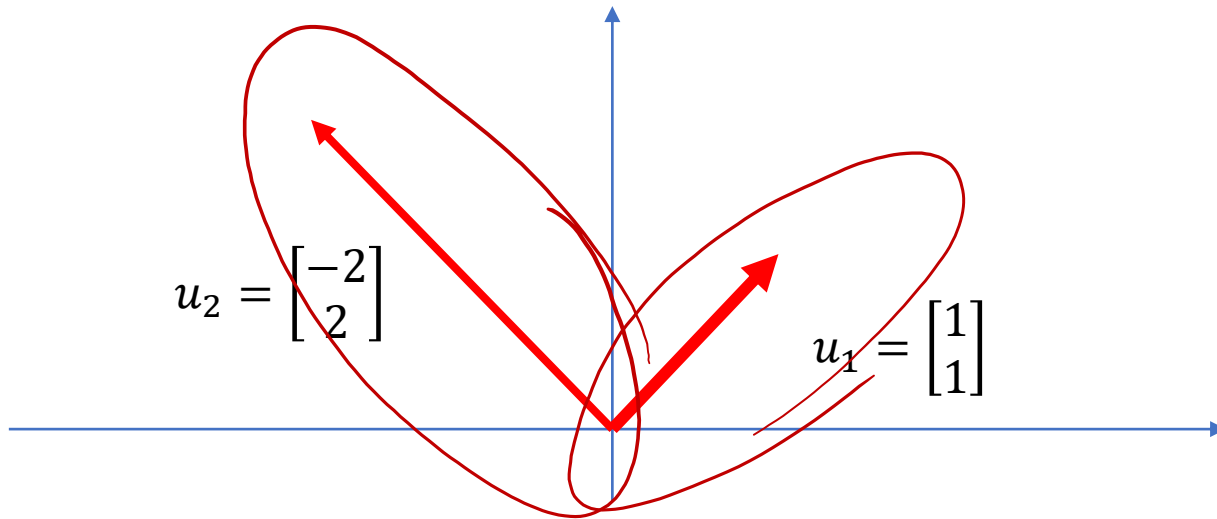
Theorem 4

If $S = \{u_1, \dots, u_p\}$ is an orthogonal set of nonzero vectors in R^n , then S is linearly independent and hence is a basis for the subspace spanned by S



Definition

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set



$$u_1 \cdot u_2 = 0$$



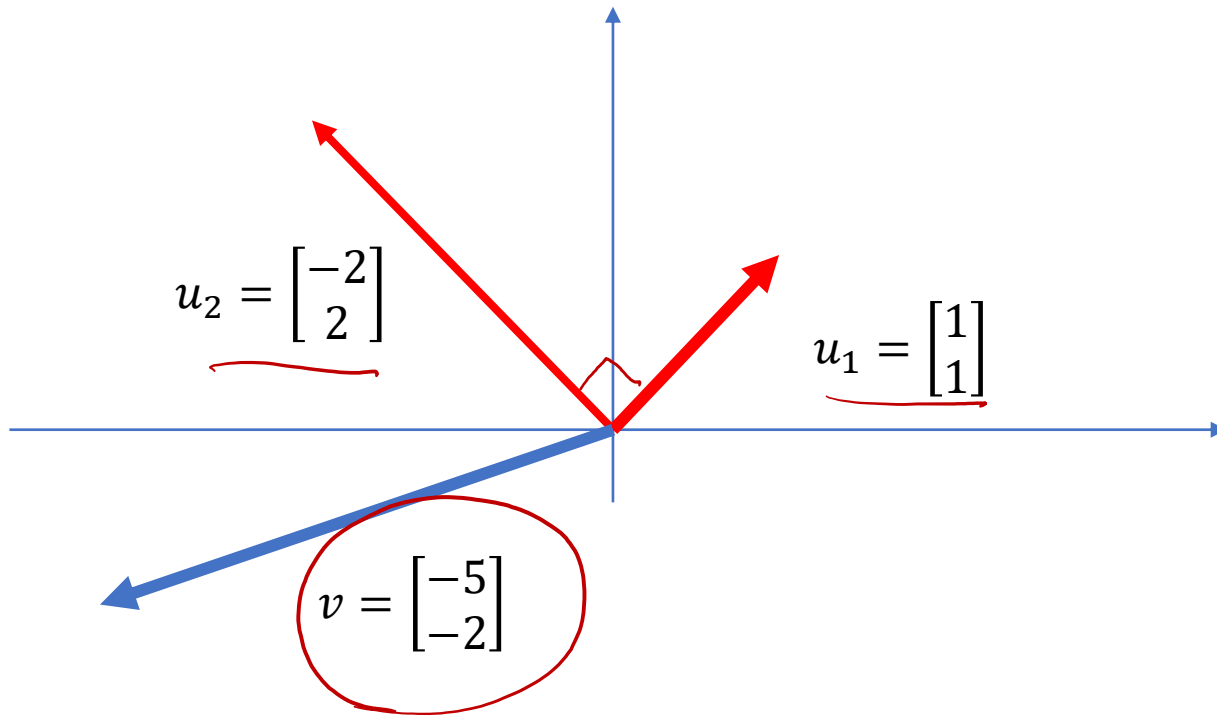
Theorem 5

If $S = \{u_1, \dots, u_p\}$ is an orthogonal basis of a subspace W of R^n ,
then for every vector w in W

$$w = \frac{\cancel{w} \circ u_1}{\underbrace{u_1 \circ u_1}} u_1 + \dots + \frac{\cancel{w} \circ u_p}{\underbrace{u_p \circ u_p}} u_p$$



Example



$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\}$$

$$v = \frac{v \circ u_1}{u_1 \circ u_1} u_1 + \frac{v \circ u_2}{u_2 \circ u_2} u_2$$

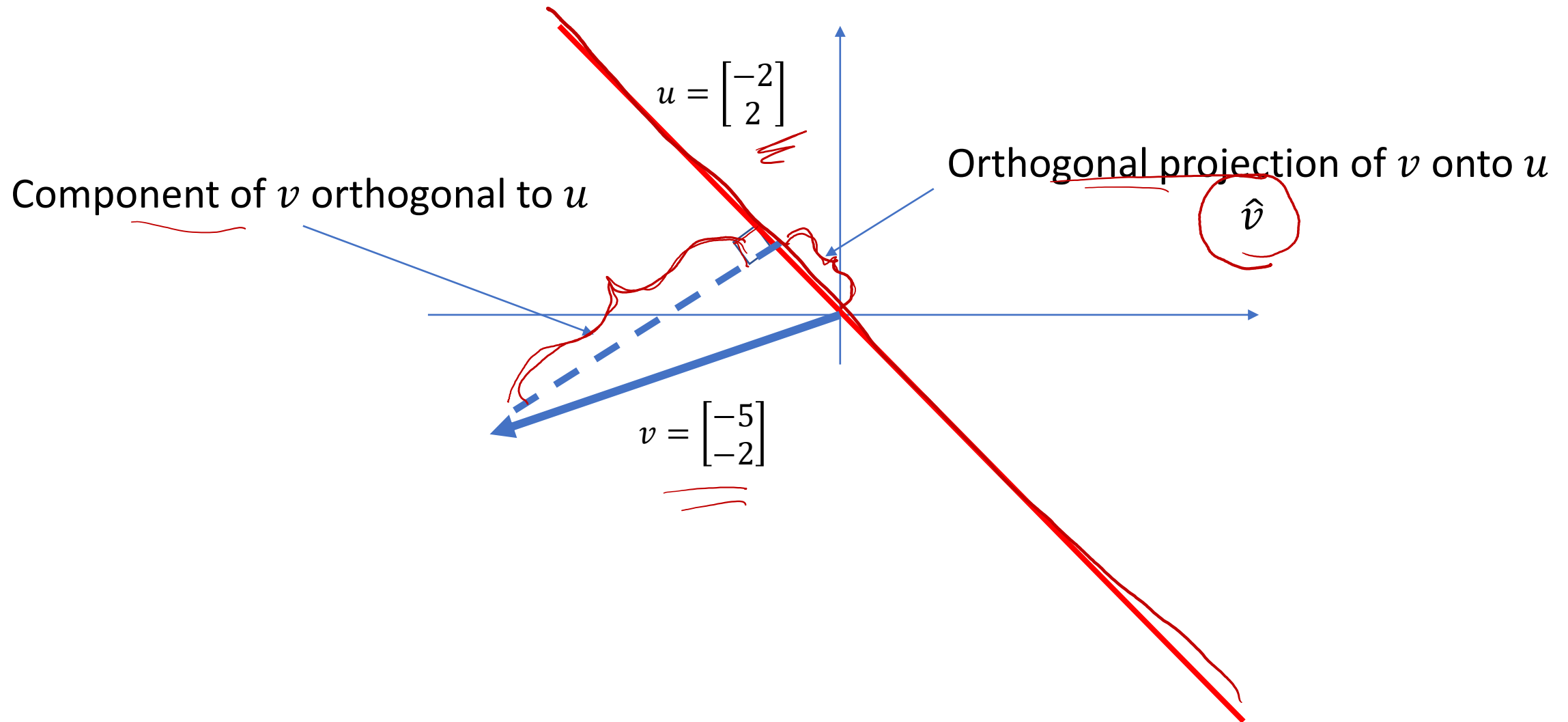
$$v = \frac{-7}{2} u_1 + \frac{6}{8} u_2$$

$$v = -3.5 u_1 + 0.75 u_2$$

$$[v]_{\mathcal{B}} = \begin{bmatrix} -3.5 \\ 0.75 \end{bmatrix}$$



Orthogonal Projection



Orthogonal Projection

The orthogonal projection of v onto u is given by $\hat{v} = \frac{v \cdot u}{u \cdot u} u$

$$v = \begin{bmatrix} -5 \\ -2 \end{bmatrix}, u = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$v \cdot u = 10 - 4 = 6$$

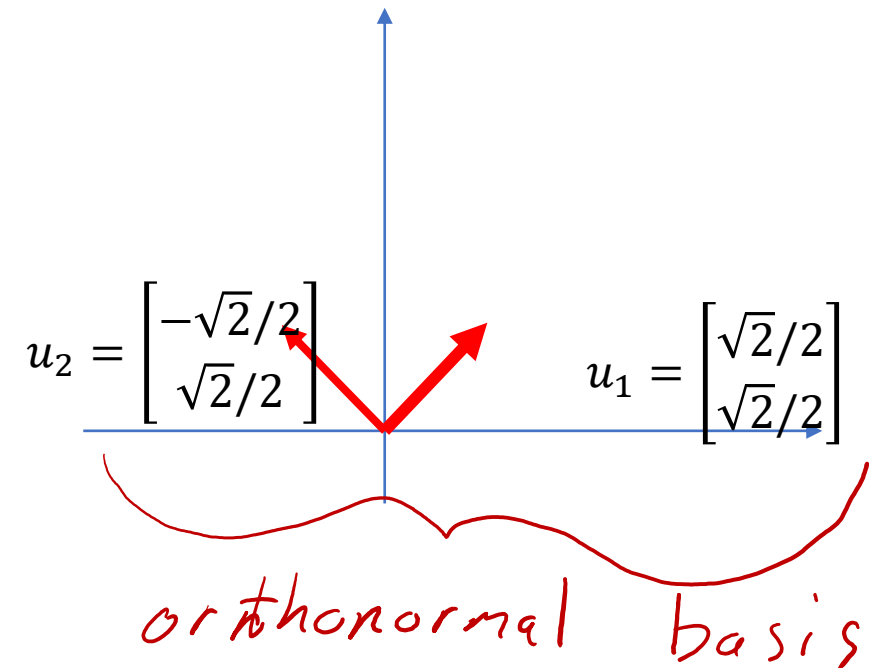
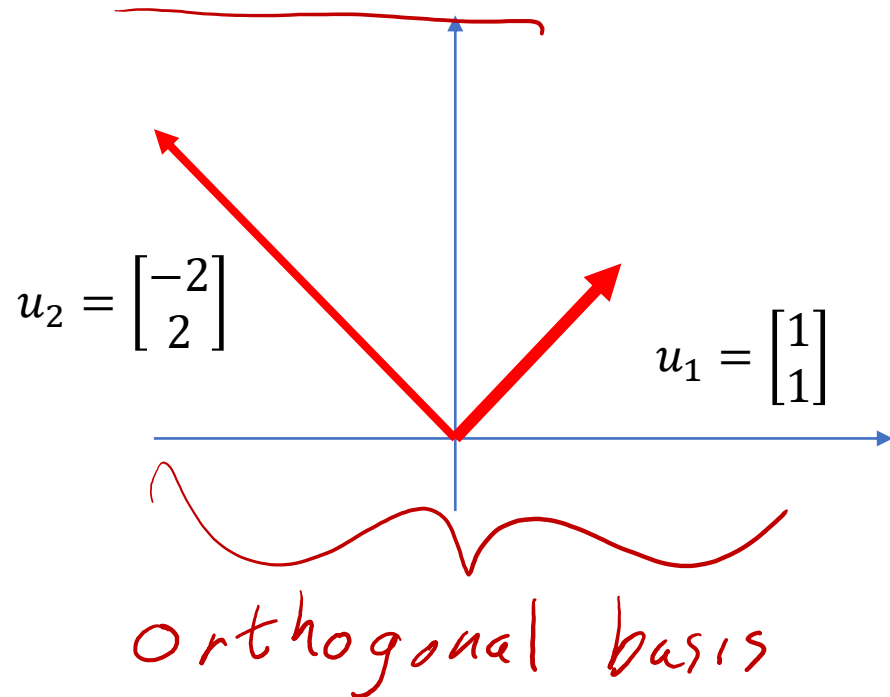
$$u \cdot u = 4 + 4 = 8$$

$$\hat{v} = \frac{6}{8} \begin{bmatrix} -2 \\ 2 \end{bmatrix} \\ = \begin{bmatrix} -1.5 \\ 1.5 \end{bmatrix}$$



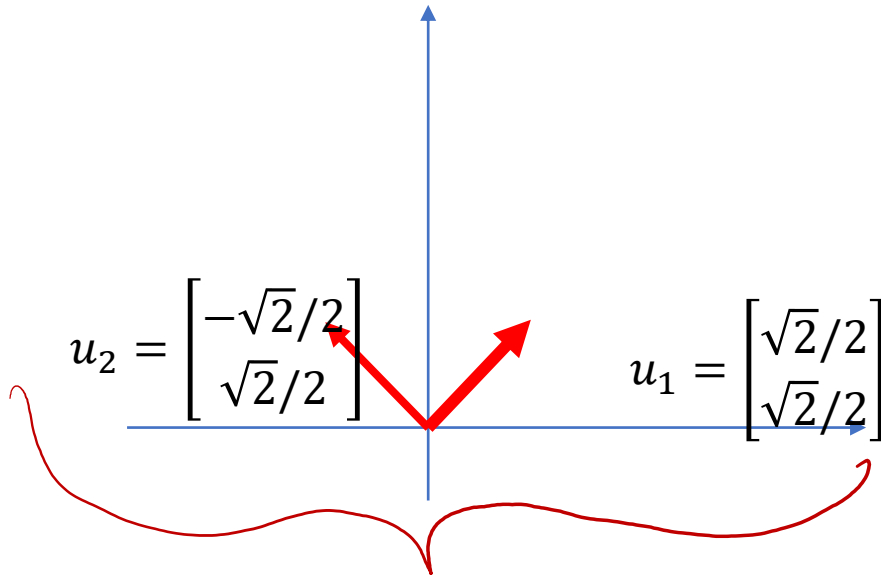
Definition

An **orthonormal set** is an orthogonal set of vectors of length 1. If W is the basis spanned by these vectors, then the vectors are an **orthonormal basis** of W



Theorem 6

The columns of an $m \times n$ matrix U form an orthonormal set if and only if $U^T \bullet U = I$



$$U = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$$U^T \bullet U = \underline{\underline{I}}$$



Theorem 7

If columns of an $m \times n$ matrix U form an orthonormal set, and x, y are vectors in R^n , then

1. $\|Ux\| = \|x\|$ 
2. $(Ux) \circ (Uy) = x \circ y$ 
3. $(Ux) \circ (Uy) = 0$ if and only if $x \circ y = 0$ 



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